

Station Labels Cannot Rank Roofs: Why Public-Data Soiling Models Do Not Transfer, and What Cleaning Is Actually Worth

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Abstract

Whether a rooftop solar array is dirty enough to be worth cleaning is measurable only from production telemetry, which vendors hold and homeowners, municipalities and service providers cannot get. We ask how far a soiling-targeting system gets on public data alone, and report where it fails and why.

Locating arrays is solved: tile-level box- F_1 **0.826** (95% CI [0.798, 0.853]), 73.6 % recall against permits the detector never saw. Ranking them is not, and our own validation hid that. Folds that hold out either *site* or *year* but not both flatter the model: ours leaves the held-out year's sites in training for **88.7%** of rows. Closing the leak moves pooled AUC from 0.710 to **0.622** (95% CI [0.571, 0.670]); a size-matched control puts 83 % of that fall on leakage (95% CI [52, 99]). Latitude and longitude alone score 0.644, and an unfitted 2006 empirical model scores 0.610.

The limit is the label unit. Public labels are measured at *stations*, so every roof in a catchment inherits one. On per-system telemetry, per-array features rank systems sharing weather at **0.573** against the **0.5** a station-label model cannot exceed by construction—but three standard cleaning assumptions put those systems' median annual loss at 2.72, 9.80 and 20.93 % and rank them at ρ 0.24 to 0.89. Per-system labels are necessary and not yet sufficient.

A third result reorders the first two, and is immune to the second. On 149 metered rooftops one wash recovers a median **\$28.10** of electricity against a \$150 service; the median roof needs **\$2.44/kWh** to break even. Annual loss cancels between the value of a clean and the recovery fraction's denominator, so the dollar figure is identical under all three assumptions while the fraction spans 0.031 to 0.217. What generalises is the fold geometry, the label-unit diagnosis and the arithmetic.

1 Introduction

Global photovoltaic capacity reached roughly 2383 GW by the end of 2025 after a record 510 GW single-year addition [12], and solar now supplies more than 8% of world electricity [11]. US small-scale systems were projected to generate about 83 TWh in 2024 [26], and the country passed six million individual installations in early 2026 [25]. Extracting output from arrays already installed is now worth as much as installing more.

Soiling, and why averages mislead.

Soiling is the accumulation of dust, pollen, and biological film on module glass. Averaged over fleets the effect is modest: Mejía and Kleissl [16] measured roughly 0.051%/day across California sites, about 7.4% by the end of a 145-day dry summer, and concluded that cleaning a typical residential system does not pay. The distribution matters more than the mean. In the same study about a quarter of sites soiled at more than twice the average, and arrays tilted below five degrees soiled roughly five times faster. Those are the systems a cleaning programme is for, and finding them is a targeting problem.

The data that would answer it is not available.

Whether a specific array is soiled is observable from its production history, which monitored inverters record and vendors retain. A production dip is not yet a cleaning decision: it does not attribute the loss to dirt, price it over a year, or weigh it against the cost of a wash. Homes without monitoring have less. Anyone outside the household, a researcher, a municipality, a cleaning service, has nothing. Today no residential array is flagged for soiling, clean or filthy.

What we found, stated before the evidence.

We built the system and it locates arrays well. It cannot rank them, and we initially believed it could, because our own validation leaked: the fold that produced our reported out-of-year AUC of 0.710 leaves the held-out year's sites in training through their other years for 88.7% of rows. Closing that leak takes the number to 0.622, and latitude and longitude on their own score 0.644—as well as all forty features. The cause is not the model. Soiling labels of the kind that exist publicly are measured at monitoring *stations*, so every rooftop in a station's catchment inherits one label, and no per-roof feature has anything to fit against.

The obvious repair is per-system labels, and we test it: on telemetry from 735 individual systems, per-array features do rank systems that share weather, at 0.573 against the 0.5 a station-label model cannot exceed by construction. But three standard ways of computing that label put the same systems' median annual loss at 2.72, 9.80 and 20.93 % and rank them at Spearman 0.241 to 0.894, and our 0.573 rests on the variant that is in the minority on ranking. The repair is necessary and not yet sufficient.

Then the result that reorders the others. We measured what a cleaning is worth on 149 metered rooftops, and one wash on the median system recovers **\$28.10** of electricity at the retail offset most of them earn, or \$10.14 at the net-billing blended rate, against a \$150 service. The median roof would need electricity at **\$2.44/kWh** to break even, about five times California retail. That is not a targeting failure and no better model repairs it. It is the more useful finding, and we put it last only because it takes the most machinery to establish.

Contributions.

We ask what a system built *entirely* on public data can and cannot establish.

1. **A gated detection stage** (§3.1; full protocol in [9]) on public county orthoimagery at 0.204 m ground sample distance, using only permissively licensed models, with an executable gate definition, confidence frozen on validation, tile-level bootstrap intervals, and an independent permit-based recall probe (§4.1).
2. **Per-array roof geometry from public lidar** (§3.1), validated against installer-reported tilt from utility interconnection filings, plus a projection trap that silently flattens tilt by 18.9 % on web-Mercator point clouds. The closest prior system, 3D-PV-Locator [15], validates tilt per system against ground truth; ours compares distributions, which is the weaker claim and we mark it as such.
3. **A measured account of why station-trained soiling models do not transfer to individual roofs** (§4.3): a leaking-fold disclosure and its correction, a variance decomposition, a controlled feature-backfill experiment, a regional holdout, and a sign reversal with an identified mechanism—together with a positive control showing that the limit is the label unit rather than the model class.
4. **A grounded cleaning-economics verdict** (§4.4): zero of 1,865 sites clear, with the one unsourced constant that had been carrying the opposite conclusion identified and replaced by a directly measured quantity: 505 observed cleaning events on 149 metered rooftops recover a median \$28.10 of electricity against a \$150 service. We show the value of a clean is invariant to the cleaning assumption while the recovery *fraction* is not, and report the fraction only as a bracket.
5. **A reproducibility posture**: every number in this paper names the artifact that produced it and the command that regenerates it (§6), and every constant we could not source is listed rather than buried (§5.5).

Figure 1 is the argument in one picture, and it is worth stating the shape before the evidence. Three of the four answers are negative, but they are not interchangeable. Question 2 fails because of a property of the labels, question 3 fails because of a property of the measurement, and question 4 fails

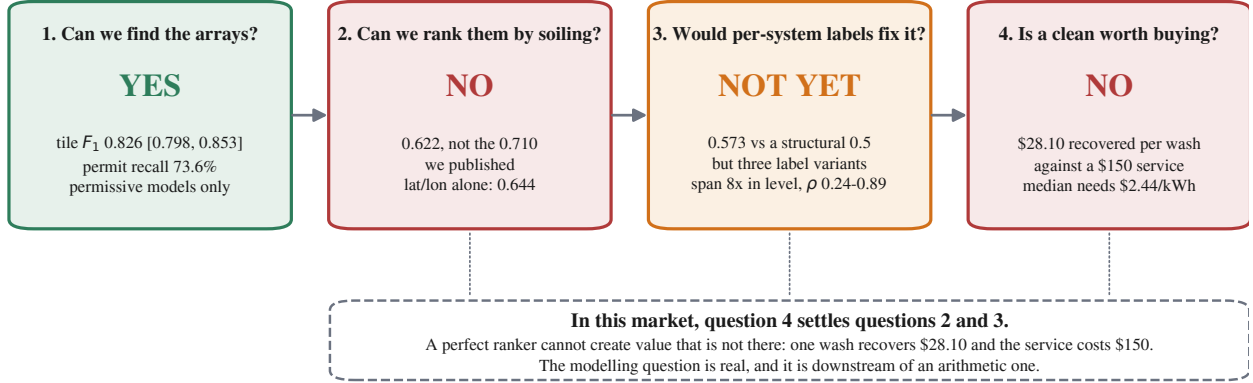


Figure 1: The whole paper. Four questions a public-data cleaning product must answer in order, what each returned, and the point at which the answer stops mattering. Question 4 is arithmetic and, *in this market*, it settles questions 2 and 3: a perfect ranker cannot create value that is not there. Where the boundary of §4.4 is cleared—drier, pricier, slower-soiling—questions 2 and 3 become live again.

for a reason that has nothing to do with either: the quantity of money involved is smaller than the cost of acting on it. A reader who takes only one thing from this paper should take the last one, because it is the only one that would not be fixed by a better model.

Figure 2 shows the pipeline end to end.

2 Related Work

Locating arrays from overhead imagery.

Public rooftop-PV detection has an established lineage: the Duke dataset [2] released 0.3 m imagery with array annotations, and DeepSolar [28] built a national deployment database by classification with weak localization. Our detector consumes that lineage rather than extending it. It uses RF-DETR [23], a real-time DETR variant [5], with an optional SAM2 [14, 22] mask stage, both Apache-2.0; the licence matters because the widely used YOLO implementations are AGPL-3.0, which obstructs reuse by exactly the municipalities and nonprofits a public-data pipeline is for. Two requirements differ from the prior work and are specified in [9]: downstream stages consume *area*, so we need instance polygons rather than image-level labels, and the gate is evaluated at the tile level through the production inference path rather than on crops. Detection is an input here, not a contribution.

Per-array geometry.

3D-PV-Locator [15] recovers tilt and azimuth per system from 3D building data and validates them per system against ground truth. We fit planes to public lidar returns inside each array polygon and can only validate distributionally, because the interconnection filings that report tilt carry no parcel, address or coordinate. We say so where the comparison appears rather than presenting the weaker design as equivalent.

Soiling measurement and modelling.

Soiling loss is extracted from yield directly by Deceglie et al. [7], and NREL publishes a station-level US soiling map [17, 18] reporting an insolation-weighted soiling ratio (IWSR). Empirical models include Kimber [13], which accumulates loss with particulate concentration and resets on rain, and SOMOSclean [17], an exponential approach to a saturation ceiling with rain-proportional cleaning. We use both as *features* under measured IWSR labels, and report in §3.2 why the converse configuration is a leakage trap we fell into and discarded. Porcar et al. [21] confirmed biofilm colonisation on panels in Berkeley but measured no power loss; that gap is the open channel of §5.2.

Uncertainty.

Our loss intervals are conformalised quantile regression [24] over gradient-boosted trees [6], because the raw quantile heads under-covered badly. Degradation and soiling-ratio decomposition use RdTools [8] on PV-DAQ [19]; cell-level occlusion physics uses the Bishop diode model [1] via pvlib [10].

3 Methods

3.1 Inputs: array detection and roof geometry

The pipeline’s first two stages supply the units this paper analyses: array polygons and their tilt. Both are specified in full in [9]. Every number this paper relies on is stated here, so nothing below depends on reading that report; it is cited for protocol detail, not for evidence.

Every input is free and public; every stage is reproducible from a committed artifact

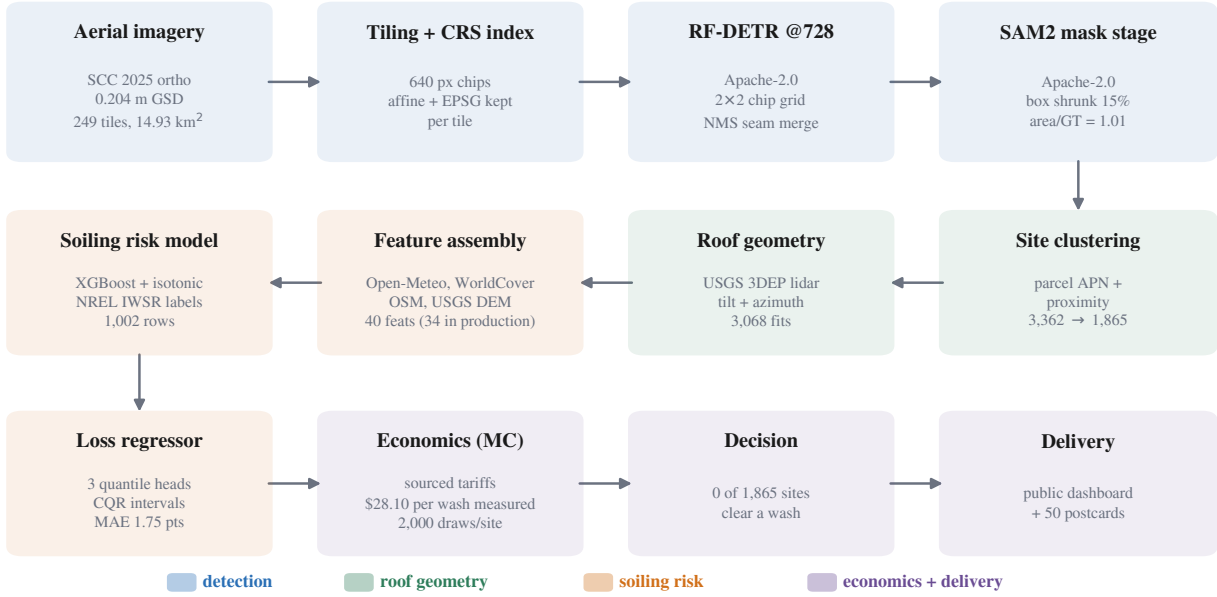


Figure 2: The pipeline. Every input is public and free: county orthoimagery, USGS 3DEP lidar, Open-Meteo weather, ESA WorldCover, OpenStreetMap, county parcel and permit records, utility interconnection filings, and NREL soiling labels. Both vision models are Apache-2.0. Stage boundaries are where the evaluation changes: detection is scored on boxes, masks on area, risk on AUC and calibration, and economics on expected net dollars.

Detection.

RF-DETR at 728-pixel input on 21 cm county orthoimagery, with an optional SAM2 mask stage, both permissively licensed. A full tile is cut into a 2×2 grid of 640-pixel chips, each chip gets one model pass, and detections are merged across the seams by NMS at IoU 0.55. The acceptance gate is stated as an executable definition rather than a number—the script is the specification—and requires tile-level box- $F_1 \geq 0.75$, a bootstrap CI lower bound ≥ 0.70 , and recall ≥ 0.70 , with the confidence threshold tuned on validation and frozen before test is scored.

Roof geometry.

Per-array tilt and azimuth are fitted to 1 m public 3DEP lidar returns inside each array polygon. One detail is load-bearing and easy to get wrong: fitting a plane to point clouds reprojected to web-Mercator reports tilt 18.9% *too shallow*, because the projection stretches horizontal distance against an unstretched vertical axis, shrinking dz/dx . Fits are performed in a local metric CRS. Arrays without a usable fit keep a flagged null rather than an imputed angle.

3.2 Soiling risk model

Labels.

We train against the NREL PV Soiling Map [18], whose IWSR is the fraction of expected insolation-weighted energy

actually delivered, so *lower is dirtier*. The annual panel release gives **891 station-year rows across 146 stations spanning 2008–2022**. A further **111 stations report only a censored summary** (IWSR > 0.99 , known-clean without an annual breakdown); we retain these at the censoring value at half the weight of a panel row, giving **1,002 rows across 257 stations**. Rows are binarized at $IWSR < 0.97 \rightarrow$ at-risk, yielding a 49.4% base rate. Panel rows span six states (AZ, CA, GA, MD, NC, OR); the full station set including summary rows spans 15. The two counts are not interchangeable.

The central methodological compromise.

We train on instrumented ground stations and apply the result to residential rooftops. That transfer is unvalidated. It is also the only option available without production data, and §4.3 measures how badly it binds.

Features.

Thirty-four features in three families, from public sources: *weather* from Open-Meteo [20] over trailing 7/30/90-day windows (precipitation, maximum 10 m wind, mean relative humidity, maximum temperature) plus two rain-timing features, consecutive dry-day streak and days since the last 5 mm rain event; *location and structure*, latitude, longitude, month, tilt, elevation, land-cover class, distance to nearest highway and to nearest agricultural parcel, and five ESA WorldCover [29] fractional covers; and *physics proxies* from

Kimber [13] and SOMOSclean [17], each as an instantaneous value plus 7/30/90-day means, with SOMOSclean’s saturation and rate re-fit to coastal California stations ($s_{\text{sat}} = 0.08$, $k = 15$).

Air quality was tested and dropped. $\text{PM}_{2.5}$ and PM_{10} rolling means covered 21% of label rows (208 of 1,002) and received zero splits in the fitted model, so the family contributed no gain at any coverage level and is excluded from the feature count above.

A leakage trap.

Physics proxies as features under measured IWSR labels are leakage-free. The converse is not. Ablations that used SOMOSclean-derived labels while retaining the SOMOSclean feature returned cross-validated AUC of exactly 1.0: the feature was the label. Those runs are discarded.

Model and validation.

XGBoost [6]: 200 estimators, max depth 4, learning rate 0.05, subsample 0.8, column subsample 0.8, L_2 regularization 5.0. Depth and L_2 were tuned against overfitting, because this many features on ~ 900 rows overfit visibly at depth 5. Outputs pass through an isotonic calibrator so the surfaced number reads as a probability. *Spatial* validation clusters stations within 10 km into the same fold. *Temporal* validation holds out each of the 15 panel years in turn and pools every row’s out-of-its-own-year prediction; §4.2 explains why this replaced a single-year holdout. *Calibration* is evaluated out-of-year by fitting each training year’s isotonic map and applying it to the held-out year.

3.3 Loss magnitude and cleaning economics

From probability to magnitude.

Multiplying a calibrated classification probability by a constant to obtain a loss percentage is a category error rather than a bad estimate. We instead use three XGBoost quantile heads (0.10/0.50/0.90) trained on $(1 - \text{IWSR}) \times 100$ under the same spatial CV and conformalised [24]. No fallback path from risk score to loss percentage exists any more, deliberately; callers without a magnitude fall back to a flagged base rate.

The decision model.

Recovered energy is priced at the site’s modelled marginal tariff and its system size, against the cost of the chosen cleaning method, over 2,000 Monte Carlo draws per site propagating the loss prediction interval, an area uncertainty ($1\sigma = 0.26$, the measured 21 cm/60 cm area shift), the rate band, and recovery timing. Economics run *per site*, not per detected polygon: 21 cm imagery fragments a roof into 1.80 polygons per site against 1.30 at 60 cm, and per-polygon costing overcharged a four-way-split roof by 2.3 $\times$.

The constant that decided everything.

`recovery_frac`, the share of a year’s soiling loss that one cleaning recovers, had been *assumed* at 0.90 and never estimated. We estimate it by simulating the soiling trajectory with and without a wash over the site’s actual rain record (§4.4).

Tariffs.

The rate had been a flat \$0.25/kWh. We replace it with a bill-reconciled PG&E E-TOU-C plus community-choice stack, validated to $\pm \$0.22/\text{month}$ over 11 consecutive real bills, giving a production-weighted retail offset of \$0.4573/kWh and a net-billing export credit of \$0.0392/kWh [4]. A soiling-lost kWh is worth the retail offset only if the home would have consumed it and the export credit otherwise, so for a net-billing customer without a battery we value it at the *net-billing blended rate*, weighting the two by a midday self-consumption share of 0.30: $0.30 \times 0.4573 + 0.70 \times 0.0392 = \$0.165/\text{kWh}$. Every “blended” figure below uses this rate, and the retail offset is the upper case. Only 5.5% of a south-facing array’s annual output lands in the 4–9 pm peak, so roughly 94% of soiling-lost kWh are off-peak. Site tariff vintage is joined from public interconnection filings [3].

4 Results

4.1 Inputs, in brief

The detection and roof-geometry stages are reported in full in the companion paper; what matters here is that the inputs to the soiling analysis are gated rather than assumed. Detection reaches tile-level box- F_1 **0.826** (95% CI [0.798, 0.853], precision 0.850, recall 0.803, $n_{\text{gt}} = 585$) at a confidence tuned on validation and frozen before test was scored, using only permissively licensed models. Recall measured independently against county building permits the detector never saw is **73.6%** on the imaged era, which is the number that bounds the funnel: a missed array is a missed roof, and no downstream model recovers it. The mask stage contributes area rather than detection, and its one dangerous failure—segmenting the whole roof instead of the array—is a prompt-box problem, eliminated at the production 15% box shrink (median mask IoU 0.844, median area/GT 1.01, roof-grab 0.0%). Per-array tilt comes from public lidar. Its validation is distributional rather than paired—the interconnection records that report tilt carry no parcel, address or coordinate, so no array can be matched to the filing describing it. The medians agree to 0.03° ($n = 3,068$ lidar fits against 8,573 filings) while a Kolmogorov–Smirnov test rejects the distributional match at $p < 10^{-90}$: every reported tilt is an integer, five values cover 57.6% of filings, and 27% are exactly 18° , the 4:12 roof pitch. Installers file a nominal pitch off a short menu; lidar measures a continuous angle. That limitation is

one of three reasons tilt is deliberately excluded from the risk model in §4.3.

4.2 Soiling risk: the gate it cleared was leaking

Which model these numbers describe.

Two feature sets appear in this work and it matters which is which. The *published* configuration carries 40 features including six air-quality columns; the *honest* production set is the same 40 minus those six, because they cover 20.8% of rows and contribute nothing measurable. Every corrected result below is reported for both, and they are indistinguishable: on the joint fold, 0.6222 for 40 features and 0.6228 for 34. We quote the 40-feature figure throughout, so that the comparison against our own published numbers is like-for-like, and give the 34-feature value wherever it differs. Where this paper says “40 features” it means the published configuration, not a recommendation.

Spatial and temporal validation.

Spatially cross-validated AUC is **0.718** (0.719 ± 0.003 over seven seeds) for the 34-feature set under the current configuration. The 40-feature set that also carried air quality reads 0.712 (0.721 ± 0.006), so dropping that family moves the metric less than the seed spread. An earlier stored figure of 0.728 for the 40-feature set no longer reproduces and should not be quoted. Against the figure we report, the margin over our 0.70 acceptance bar is under two points, and against the 40-feature set it is one.

Our initial temporal design held out 2022 as a single test year and returned AUC 0.679 on 97 rows. That estimate is unusable, and we report it because it is the kind of number over-read in both directions: its Hanley–McNeil standard error is 0.061, its 95% CI is [0.557, 0.797], and the 0.70 bar sits 0.32 standard errors away. A 97-row year cannot distinguish 0.68 from 0.70. Replacing it with rolling leave-one-year-out pooling over $n = 891$ (440 at-risk, 451 not) gives AUC **0.710**, SE 0.017, bootstrap 95% CI [**0.676**, **0.742**], $P(\text{AUC} \geq 0.70) = 0.70$: a $3.5\times$ tighter estimator on identical data. **That estimator is also leaking, and the leak is the finding.** Leave-one-year-out holds out a year but keeps the held-out year’s sites in training through their other years: measured, **88.7%** of held-out rows have their own site in the training set. The 10 km spatial fold has the mirror defect, grouping by location while mixing years. Neither is a test of a roof we have not seen in a year we have not seen. Building that fold—hold out year y , then drop from training every row whose site appears in y —gives AUC **0.622**, 95% CI [**0.571**, **0.670**], $P(\text{AUC} \geq 0.70) = \mathbf{0.001}$, on the same 891-row pooled basis. *The model does not clear its temporal gate.* Latitude and longitude alone score 0.644 on that fold, at or above the full 40-feature set, which is the identifiability problem appearing directly in the metric: station-level labels

give every roof in a catchment one label, so no per-roof feature has anything to fit.

What the prior literature gets, scored the same way.

The natural next question is what a published soiling model achieves on this task. Two are already computed over our rows—Kimber [13] and SOMOSclean [17]. Kimber is used exactly as published, with no parameter estimated on our data, so it needs no fold: there is nothing fitted to leak. SOMOSclean is *not* in that position, and we flag it rather than lean on it—its saturation and rate were re-fit to coastal-California stations (§3.2), some of which are rows scored here, so its 0.544 should be read as an optimistic bound on an unfitted SOMOSclean rather than as a clean baseline. Kimber carries the comparison. Scored on the same 891 panel rows, Kimber’s instantaneous IWSR proxy gives AUC **0.610** (95% CI [0.566, 0.652]) and SOMOSclean’s gives 0.544. Handing all seven physics columns to the learner under the joint fold gives 0.5706.

Set that beside our own numbers on the identical basis—0.622 for forty engineered features, 0.644 for latitude and longitude—and the ordering is the result. A 2006 empirical formula with no fitting is within noise of a gradient-boosted model trained on fifteen panel years of weather, air-quality and static features, and two coordinates match or beat both. This is the same conclusion the feature ablations reach, arrived at from outside: on station-level labels there is very little to learn beyond where the station is.

Is the drop the leak, or the smaller training set?

Closing the leak also deletes data: holding out year y and then removing every row whose site appears in y costs a median 28.8% of the leave-one-year-out training set, and 74.3% in the worst year. Some of the fall could therefore be volume rather than leakage, and the control is cheap. For each year we build a third training set of the *same size* as the joint fold’s that *keeps* the leak: start from the leave-one-year-out set and delete an equal number of rows drawn only from sites absent from year y . Such deletions cannot remove a leaking row, because the leaking rows are exactly those from sites present in y .

Training set	pooled AUC
leave-one-year-out, leak present, full data	0.7144
size-matched control, leak present	0.6986
joint out-of-site and out-of-year	0.6222

Of the 0.0922 total, **0.0764 (83%) is attributable to the leak and 0.0158 (17%) to training size.** Bootstrapping the decomposition over stations puts the leak share at 95% CI [52, 99]% and the leak’s own contribution at [+0.031, +0.121], which excludes zero. The split is therefore not precisely resolved at this sample size: what the data support is that the *majority* of the fall is leakage, not that the share is 83%. The five-seed spread on the control itself is

0.0012, so the imprecision is sampling variance in the AUCs rather than which rows were deleted.

That 0.622 is not a fragile number. Across seven seeds it is 0.6201 ± 0.0047 ; penalised logistic regression, a different model class entirely, gives 0.5707, which is lower rather than higher; and the weighting of the censored summary rows—the choice a referee is most likely to query—moves it only between 0.594 (dropped) and 0.622 (production). No setting we tried approaches the 0.70 bar. The regional result is equally insensitive to fold geometry, ranging 0.638 to 0.664 as the number of k-means regions varies from 5 to 12.

We report the leaking numbers alongside the corrected ones because reproducing them requires knowing which of three reporting bases is in use, and none was previously written down: pooled against mean-of-folds (about 1 point), all 1,002 rows against the 891 panel rows (3 to 6 points, since the 111 censored summary rows are easier to classify and all carry a single sentinel year), and a plain gradient-boosted fit against the calibrated training pipeline (about 0.4 points). Matched on basis, every previously published figure reproduces to within 0.012. Out-of-year calibration is retained: Brier 0.218 against a base-rate reference of 0.250, ECE 0.046, MCE 0.126 (Figure 3). The weakest year is 2019 (AUC 0.626); 2008–2012 are near single-class.

4.3 Why the model cannot rank individual roofs

This is our second negative result, and it reframes what the pipeline claims. The model is valid for *level* and not for *ranking homes against each other*.

Variance decomposition.

Across the 1,710 residential-scale sites (2–15 kW), the per-home dollar figure is driven by system size (p10–p90 ratio $3.20\times$, **86.7%** of variance) and roof orientation ($1.30\times$, **11.1%**); the soiling model contributes $1.15\times$ and **2.2%** (Figure 4). **97.8% of the spread comes from what the detector and the lidar measure.** Doubling the soiling model’s spread would add less differentiation than joining one public permit file. The decomposition mixes per-array and per-site quantities, so it estimates rather than partitions; the ratio is scale-invariant and therefore unaffected by the level calibration below.

The mechanism.

NREL labels are station-level. Across a 20 km coastal AOI, feature standard deviations are 0.3% to 23% of the training standard deviations, and **58.1% of model importance sits on features effectively constant across the AOI.** A model trained to separate 15 states has almost nothing left when shown 1,865 homes in one town, because the training set contains *zero* examples of per-roof variation.

The controlled experiment.

Several features were missing from the deployment matrix and silently median-filled, an obvious candidate explanation with an obvious fix. We supplied the WorldCover features properly, resolving all 3,349 points. The within-AOI p10–p90 spread moved from 0.776 to 0.758 points, which is to say not at all, and dead importance moved 64.9% to 58.1%. The plumbing gap was not the cause.

The sign reversal.

Arrays on tree pixels scored *lower* (5.36%) than built-up (5.55%). The model learned `worldcover_tree` from NREL stations, where a tree pixel means “forested rural region, wetter air, less dust,” not “a tree overhanging this roof.” **Same feature name, different referent.** We predicted and then confirmed the same failure for tilt, and deliberately excluded our measured per-array tilt from the risk model for three measured reasons: the training interquartile range $[7.49^\circ, 12.00^\circ]$ contains only 8.9% of our roofs; gradient-boosted trees do not extrapolate, so the deepest tilt split is at 26.0° and predictions at 30° , 35° and 40° are identical; and the training feature is effectively a station identifier (42.4% of rows at exactly 7.49° , 24 unique values, 0.372 correlation with longitude). The learned effect accordingly carries the *wrong sign*. Tilt belongs in the physics, where a 30° roof self-cleans on less rain than a 5° one, not in a model that never saw per-roof tilt vary.

The mechanism, measured on metered roofs.

Figure 5 shows what the label unit costs. One 0.5° cell in our fleet holds 103 metered residential systems whose measured annual soiling losses range from 0.23 to 5.35 %, a factor of 23. A station-level label assigns all 103 the same number. That cell is not a special case: across the 37 cells holding at least five metered systems, the within-cell standard deviation is a median 42 % of the cell mean. Approximately half of the variation a per-roof model would have to explain is removed before training begins, and no feature engineering reaches it.

What per-system labels recover.

Everything above is a diagnosis without a remedy, and a reader is entitled to ask whether *any* public-data model can rank roofs or whether the task is impossible. It is not. The decisive test is ranking systems that share weather: two arrays in the same 0.5° cell see the same climate, air quality and rainfall, so a station-label model assigns them identical features and emits identical predictions. Its within-cell pairwise concordance is therefore exactly 0.5 *by construction*, not as an empirical finding. Anything above 0.5 must come from per-array information.

We test this on the PVDAQ fleet, where the labels are per system rather than per station: 94 cells holding two or more systems, 735 systems, 11,653 same-cell pairs, evaluated leave-one-cell-out so the model never sees the cell it

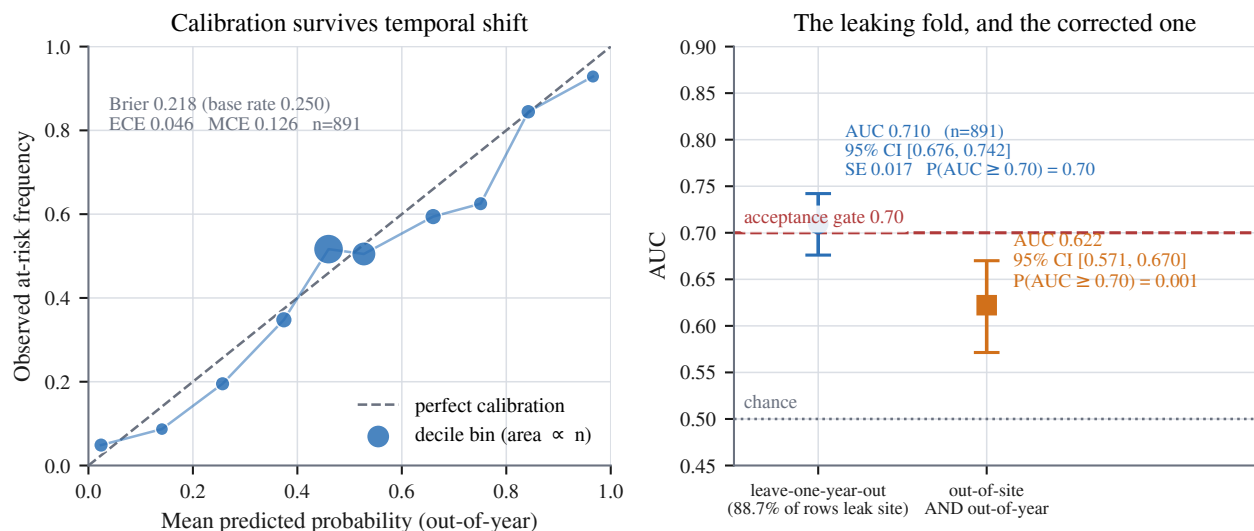


Figure 3: Out-of-year reliability, and the two folds side by side. *Left:* calibration survives temporal shift. *Right:* the published leave-one-year-out interval, whose fold leaves the held-out year’s sites in training for 88.7 % of rows, against the same model scored jointly out-of-site and out-of-year. The published interval straddles the acceptance bar; the corrected one sits below it, with $P(\text{AUC} \geq 0.70) = 0.001$.

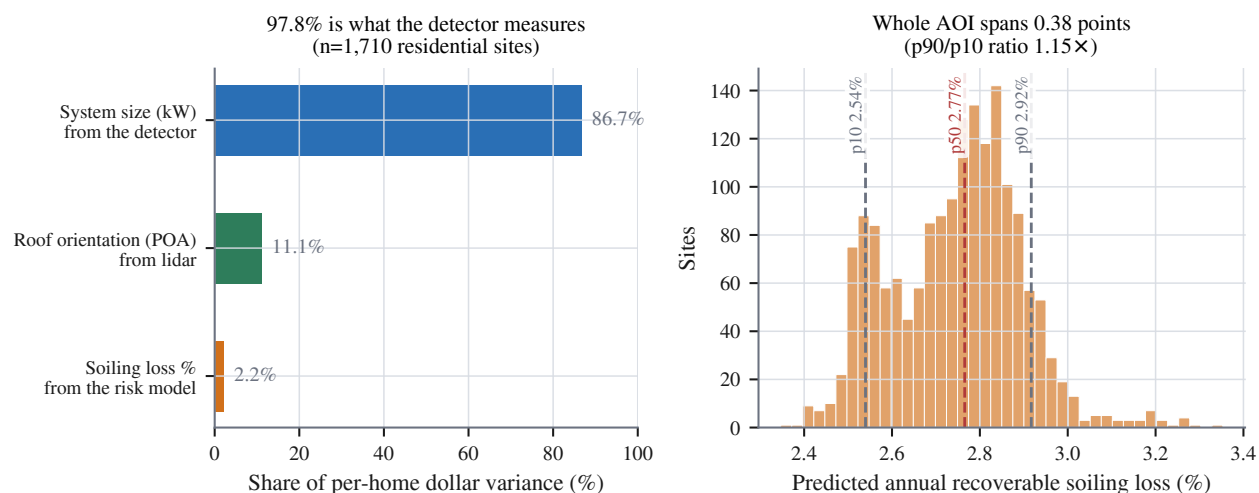


Figure 4: Left: share of per-home dollar variance by driver. Right: the soiling head’s entire within-AOI spread, 0.38 points p10–p90 across 1,710 homes.

ranks (Figure 6). Per-array features alone—tilt, azimuth, capacity, age—give concordance **0.573** (95% CI [0.512, 0.609], $P(> 0.5) = 0.995$, bootstrapped over cells because systems within a cell are correlated). Location alone, the station-model analogue, gives **0.484**, indistinguishable from chance even with sub-cell coordinates available. Per-array information is therefore present and learnable, and the station-label ceiling is structural rather than incidental.

But the per-system label is not yet well posed, and this is the more useful finding.

The result above uses the `perfect_clean` soiling ratio. RdTools offers two further cleaning assumptions, and

on 146 California residential systems where all three fits are non-degenerate they disagree about both the level and the ordering:

assumption	median loss (%)	IQR (%)	median CI width (pts)
perfect	2.72	[1.95, 3.74]	0.63
half_norm	9.80	[8.53, 11.98]	6.92
random	20.93	[17.34, 24.33]	19.84

Spearman between them, on these 146 systems with all three fits non-degenerate and a tighter Monte Carlo: perfect-half_norm **0.241**, perfect-random **0.318**,

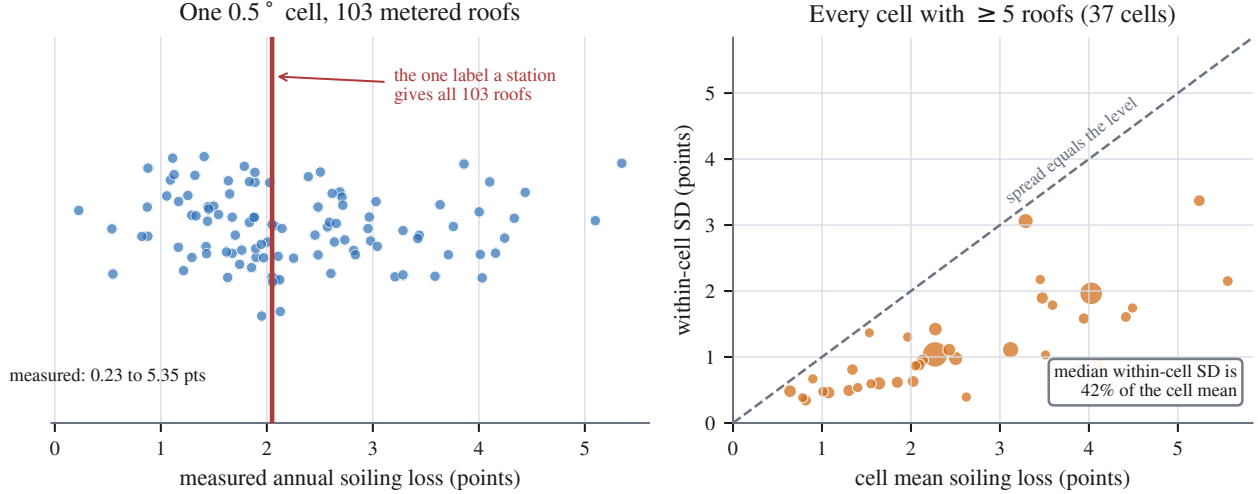


Figure 5: The identifiability problem, measured rather than argued. *Left:* every metered residential system inside one 0.5° cell. Their annual soiling losses span 0.23 to 5.35 %, and a station-level label gives all 103 of them the single value marked in red. *Right:* the cell is not unusual. Across all 37 cells holding five or more metered systems, the within-cell standard deviation is a median 42 % of the cell mean. Roughly half the variation a per-roof model would need to explain is variation that station labels discard by construction.

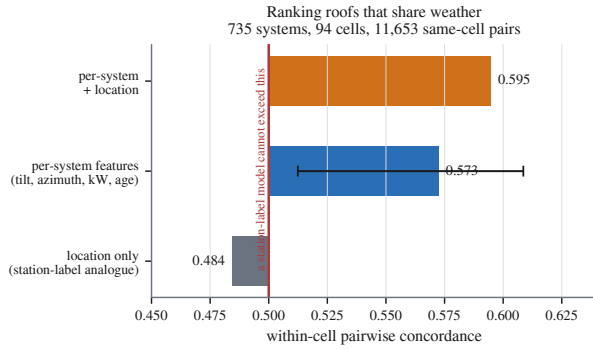


Figure 6: Within-cell pairwise concordance on systems that share weather. A station-label model cannot exceed 0.5 here by construction, and per-array features clear it on `perfect_clean` labels. The result does not survive the alternative cleaning assumption (`half_norm_clean`, concordance 0.512), which is why we report the label instability rather than the ranking as the finding.

and `half_norm-random` **0.894**. Re-running the within-cell experiment against `half_norm_clean` gives concordance **0.512**, at the edge of chance. (The `perfect_half_norm` correlation is 0.126 across the larger 735-system ranking sample and 0.241 here; the two populations and Monte Carlo budgets differ, and we quote 0.241 throughout because it is the run in which all three assumptions were computed on identical systems.)

This is not one bad estimator. That was our first reading and it does not survive the third assumption: `half_norm` and `random` are different assumptions and they rank the same roofs at 0.894, while `perfect_clean` disagrees with both. The uncomfortable part is that the plausible *level* and the majority *ranking* point at different estimators. Only

`perfect_clean` lands near the independent references for this population—2.76 % for PVDAQ systems near the pilot, 4.70 % for NREL’s coastal-California stations—and that comparison is partly circular, because NREL’s IWSR is itself defined under a perfect-cleaning assumption. Our own 0.573 therefore rides on the minority ranker, and we say so rather than let a reader discover it.

The honest statement is narrower than the section began. Per-array signal exists, per-array covariates exist, and station labels can reach neither. But a per-system label recovered this way is *necessary and not yet sufficient*: the choice of cleaning assumption moves the level eightfold and can move the ranking to chance, and no independent reference can adjudicate it, because the references available share an assumption with one of the candidates. Closing that gap—an estimator both unbiased for recoverable loss and tight enough to order two roofs in one neighbourhood—is the prerequisite for per-roof soiling targeting. It is a measurement problem, and it is the one worth working on. We flag it because the obvious next move in this literature is to scale per-system label extraction, and scaling a label whose standard variants disagree this much will produce confident and unreproducible rankings.

The dataset is less geographically balanced than a station count suggests.

Counted over all 257 stations, 57 % sit west of -114° , which sounds tolerable. Counted over the 146 stations that contribute *panel* rows—the only ones with multiple years, and therefore the only ones a temporal fold can use—the figure is 91 % (Figure 7). The eastern representation is almost entirely the 111 censored summary rows, 87 % of which are eastern and each of which contributes a single observa-

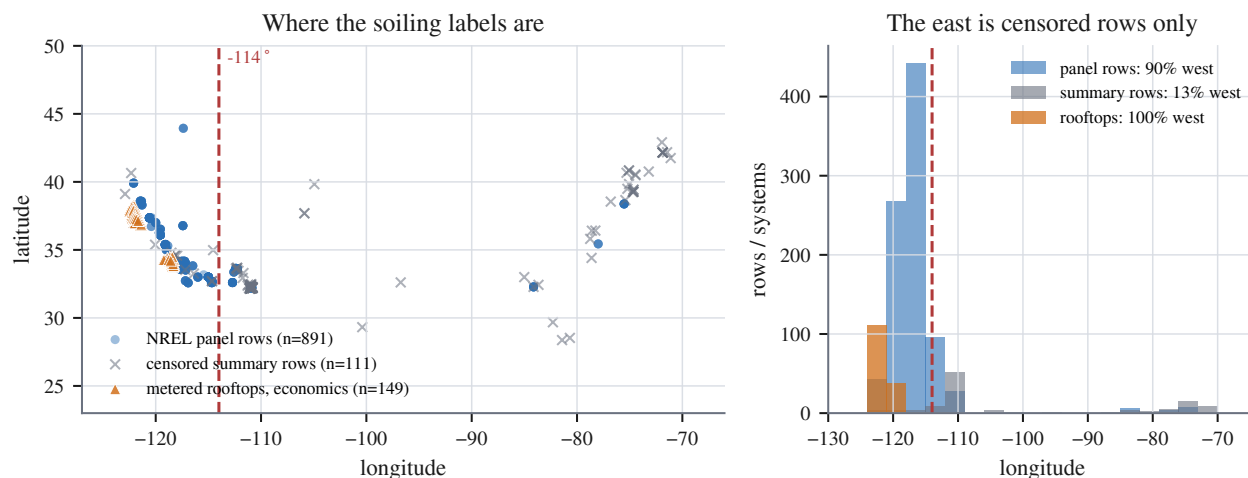


Figure 7: Why no model here should be expected to travel. The 891 annual panel rows that carry temporal structure are 89.6 % west of -114° , and 91 % of the 146 stations behind them. The eastern United States appears in this dataset almost entirely as *censored summary rows*—one row per station, no annual breakdown, 87 % of them eastern—which contribute nothing to a fold that holds out a year. Every metered rooftop in our economics is Californian.

tion with no annual structure. A geographic holdout on this dataset is therefore closer to a within-Southwest holdout than the station count implies, and we read our own out-of-region numbers accordingly.

Two caveats on those two numbers, both of which we found by re-running the experiment. The fold holds out whole sites, since a region contains them entirely, but it does not hold out years; and the script read its hyperparameters from a configuration key that does not exist, so it had always fitted a stock model rather than the one we ship. Correcting both gives pooled **0.655** and **0.527** on the largest region. The direction of the conclusion is unchanged and its magnitude is worse. 89.6 % of the panel rows sit west of -114° , so the model has effectively already seen the Southwest. This model does not travel to an unseen region. More features will not fix it; a geographically balanced label set will.

Level is biased high, and we correct the symptom, not the cause.

The raw head predicts 5.53% annual recoverable loss for this AOI. PVDAQ residential systems within 120 km measure a median of 2.76% ($n = 118$). We apply a multiplicative calibration of 0.500 to all three quantiles, placing the AOI at 2.76% against NREL’s own coastal-California station median of 2.80%, an independent reference not used to derive the factor. The cause is the training geography: 774 of 891 rows are California, dominated by Los Angeles and San Bernardino, which is desert. The fix is a per-array label set, not a scalar.

4.4 The cleaning-value verdict

Zero of 1,865 detected Santa Cruz sites show a positive expected net from a cleaning, on recoverable soiling, at any

electricity rate to \$0.70/kWh, at any system size, over 2,000 Monte Carlo draws each. Maximum $P(\text{net} > 0)$ across all sites is 0.0000. Figure 11 shows the shortfall, cleaning cost divided by annual recovered dollars, for every site.

The same verdict on measured rooftops, with nothing assumed.

The result above rests on a modelled recovery fraction and a detected, not metered, array. We can now remove both. For **149** residential California systems in PVDAQ we measure recovery directly from RdTools soiling intervals—505 cleaning events falling in dry windows, separated from rain resets by daily precipitation—together with each system’s own annual soiling loss, its observed heavy-rain frequency, and its nameplate capacity. *What we measure is the value of a clean, not a recovery fraction.* One wash recovers a median **241 point-days** of performance (IQR [160, 340]), worth **\$28.10** at the retail offset and \$10.14 at the blended rate (IQR [17.32, 43.70] and [6.25, 15.77]), against a median \$150 service. Those are assumption-free: they depend on the observed step and its duration, not on any model of unobserved cleaning.

The dollar verdict does not depend on any of that.

§4.3 shows the per-system *label* is assumption-dependent by a factor of eight. It would be reasonable to expect the economics to inherit that, and they do not, for an algebraic reason worth stating explicitly. Recovered dollars are the annual loss in dollars multiplied by the recovery fraction, and the recovery fraction carries that same annual loss in its de-

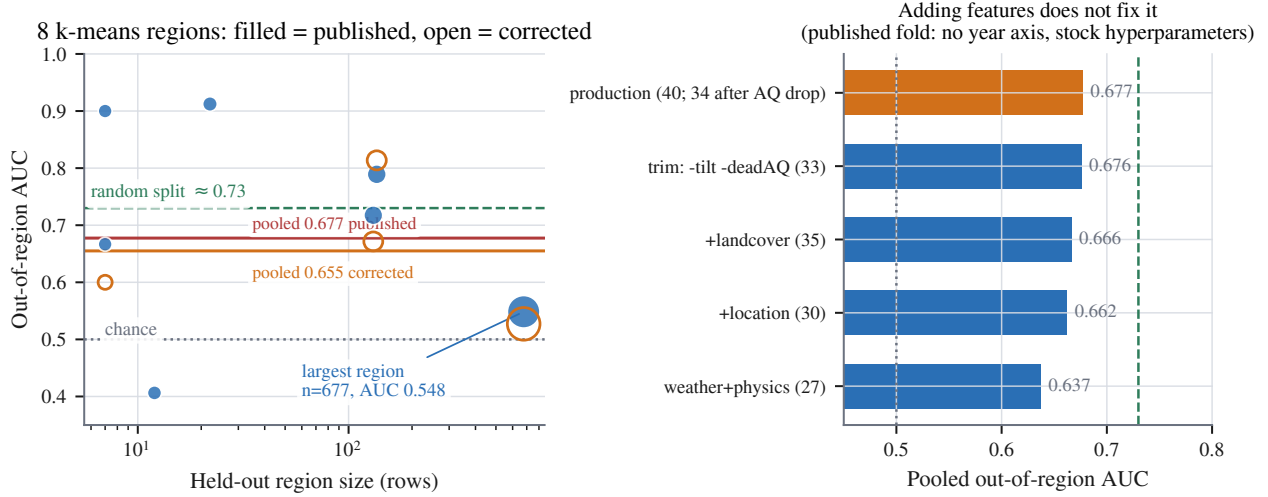


Figure 8: Out-of-region generalization. The largest held-out region scores near chance, and no feature set recovers it.

nominator:

$$\underbrace{kW \cdot h \cdot \eta \cdot p \cdot \frac{L}{100} \cdot 365}_{\text{annual loss in \$}} \times \underbrace{\frac{\Delta \cdot d}{\frac{L}{100} \cdot 365}}_{\text{recovery fraction}} = kW \cdot h \cdot \eta \cdot p \cdot \Delta \cdot d.$$

The annual loss L cancels exactly. What a clean is worth depends only on the measured performance step Δ , how long it persists d , the system size, the insolation and the tariff—none of which involves an assumption about unobserved cleaning. Verified on every system rather than one: across all 149 rooftops and all three cleaning assumptions, spanning an eightfold range of assumed annual loss, the largest deviation in recovered dollars is 4.3×10^{-14} dollars, which is floating-point noise. *The contribution most exposed to the label problem is the ranking; the one that decides the product is immune to it.*

Recovery fraction is a ratio, and we demote it accordingly.

The literature’s constant—assumed at 0.90, modelled here at 0.045—is $\Delta d/L$, a measured numerator over an *assumption-dependent* denominator. By the same cancellation that makes the dollars robust, the ratio inherits the full label instability of §4.3. On the 112 systems carrying all three cleaning assumptions:

denominator	median L	implied recovery fraction
perfect_clean	2.97 %	0.217 [0.132, 0.361]
half_norm_clean	9.61 %	0.063 [0.048, 0.097]
random_clean	20.58 %	0.031 [0.022, 0.044]

A sevenfold range, entirely from the denominator. We therefore do *not* report a single measured recovery fraction, and an earlier draft of this paper was wrong to. What survives is one half of a bracket: the **0.90 originally assumed is above**

all three, so the original business case was wrong by more than the uncertainty in any direction. The modelled 0.045 is *not* below all three—`random_clean` implies 0.031—so it falls inside the measured range rather than beneath it, and we cannot claim the simulation is conservative against every denominator. Anyone recomputing from the published fleet label file obtains 0.217, that file carrying `perfect_clean`. At the net-billing blended rate of \$0.165/kWh, **zero of 149** systems clear; at the full retail offset of \$0.4573/kWh, two do. The median system requires **\$2.44/kWh**, roughly five times California retail (Figure 9, left). Peak-sun-hours are measured per site from the cached irradiance record rather than held at the 5.5 h constant the pipeline previously applied everywhere; that constant is a Santa Cruz value, the measured median across 29 cells is 5.09 h, and the constant is too high for 22 of them—so it had been making the cleaning case optimistic rather than conservative.

The obvious objection is our \$150 service floor, which is unsourced and which 135 of the 149 systems price at. It is not load-bearing, because what a wash returns is too small for any cost structure to reach. One wash on the median system recovers **\$28.10** at the retail offset and \$10.14 at the blended rate (Figure 9, right). A wash would have to cost under \$10 before half of these systems cleared at the blended rate; at \$25, below any plausible service price, 9.4% clear.

This comparison is per-wash on both sides, which is what makes it robust. A homeowner buys one wash at a time, each worth Δd priced, and by the cancellation above that value carries no cleaning assumption. We previously described the same figure as “the entire annual recoverable value”, which was a mislabel rather than a different number: the two differ only by using the 5.5 h constant instead of each site’s measured insolation, and agree to 4×10^{-14} dollars per system when computed alike. An annual prize *would* depend on the cleaning assumption, and we make no claim about one.

Where cleaning would pay.

Stating where the product fails is weaker than stating where it would work, so we sweep the boundary (Figure 10). Rain frequency enters twice and both times against the seller: more rain means less accumulated loss to recover, and a shorter benefit window before the next natural reset. Both quantities come from the same daily trajectory, so they compound rather than multiplying as independent constants. Reading the boundary off the full grid, a positive expected net requires *all* of a re-soiling time constant $k \geq 30$ d, at most 15 heavy-rain resets per year, a tariff above \$0.20/kWh and a system above 4 kW. Each threshold is the most permissive value at which any grid cell clears while the other axes are free, so they are necessary conditions read jointly, not a recipe: only 323 of 49,920 cells are positive, and Figure 10 plots one deliberately generous slice of that grid rather than the thresholds themselves. The surface is non-monotonic in k : too fast and there is no benefit window, too slow and the array never soils enough to be worth cleaning. Coastal California sits at $k = 15$ d with 16.5 resets per year, far inside the losing region.

Scope of the verdict.

IWSR is defined as the insolation-weighted mean of daily soiling ratios *assuming perfect cleaning*, so a permanent wash-only layer is excluded from the labels by construction. Our loss model trains on $(1 - \text{IWSR}) \times 100$ and therefore predicts *recoverable* loss, not total output deficit. Everything in this subsection is silent about the other channel (§5.2).

The constant that was carrying the product.

`recovery_frac` was assumed at 0.90 and models at **0.045**, a 20 $\times$ overstatement. The 0.045 is a simulation output, not a measurement: it depends on a fitted re-soiling constant and on the SOMOSclean saturation whose fit we qualify in §5.5. The mechanism: at the calibrated re-soiling constant $k = 15$ equivalent-days an array returns to its annual mean in about two weeks, and Santa Cruz gets 27 heavy-rain (≥ 10 mm) resets a year for free. One cleaning buys roughly one array-month of clean out of twelve. Two validations: the same trajectory reproduces NREL’s measured coastal-California annual loss (model 5.06% against measured 4.70%), and the result is robust to k —even at $k = 60$, four times the calibrated value, in Phoenix, recovery is only 0.125. **With 0.90 restored, 44.8% to 95.4% of the AOI “should clean.”**

A counter-correction that does not move the verdict.

Pricing every site at the current net-billing tariff was itself an untested assumption. Public interconnection records [3] filtered to the five AOI ZIP codes ($n = 7,536$ residential PV) show **90.1% of the AOI on legacy net metering** (52.5% NEM 2.0, 37.6% NEM 1.0). The population-blended value of a lost kWh rises from \$0.1646 to \$0.4284, a 2.60 $\times$ understatement, and the AOI annual recoverable loss total rises

Table 1: Shortfall (cleaning cost $\div$ annual recovered dollars) by system size. Values > 1 mean the wash costs more than it returns. Computed from `site_economics.csv` with the *modelled* recovery fraction of 0.045 and the level calibration applied. §4.4 shows that fraction is denominator- dependent (0.031 to 0.217) and reports the value of a wash in dollars instead; the shortfall ordering below is unaffected.

Size band (kW)	n sites	Median	Best
0–5	877	45.9 $\times$	26.5 $\times$
5–10	796	24.6 $\times$	13.8 $\times$
10–15	119	16.6 $\times$	13.7 $\times$
15–30	35	13.8 $\times$	11.2 $\times$
30–60	16	10.4 $\times$	8.6 $\times$
60–150	12	8.5 $\times$	6.7 $\times$
150+	10	5.5 $\times$	3.6$\times$
All	1,865	—	0 clear

156%. **The verdict did not move:** still zero of 1,865, still maximum $P(\text{net} > 0) = 0.0000$.

Where the gap sits.

Table 1 reports the shortfall by size band. Targeting by size buys roughly 8 $\times$ and then asymptotes near 5 $\times$, because recovered dollars and cleaning cost both scale with panel count. Small bands look worse only because the minimum service charge dominates a twelve-panel job. Re-pricing across the published California market range, \$5 to \$20 per panel, **zero sites pay at any price on the board.**

4.5 Pilot deployment

The pipeline ran end to end on the AOI, producing 3,362 detected arrays resolving to 1,865 sites, a public dashboard with per-array size, orientation, and modelled loss, and 50 postcards rendered and mailed live on 2026-06-30 (50 of 50 accepted, \$47.52, QR deep-links verified for all 50 identifiers). We present the mailing as a deliverability and mechanics test, not as a behavioural experiment: a 50-card drop cannot resolve a response rate statistically. The dashboard’s honest claim is a per-home diagnosis: this is your array, this is its size and orientation, this is what it loses, and no, do not pay to clean it.

Figure 12 is that diagnosis, captured before the level calibration described above. We show it because the claim “the honest deliverable is telling someone not to buy” is otherwise unverifiable, and because the panel carries the paper’s own caveat into the product: it tells the homeowner that the figure assumes a best-week clean, that an average date recovers about a third as much, and that a second soiling channel exists which nobody has measured here.

In the same footprint the previous-generation stack, built on 0.6 m NAIP imagery [27], found 334 arrays. Polygon counts are not comparable across GSD (1.80 polygons per site at 21 cm against 1.30 at 60 cm), sites are the comparable

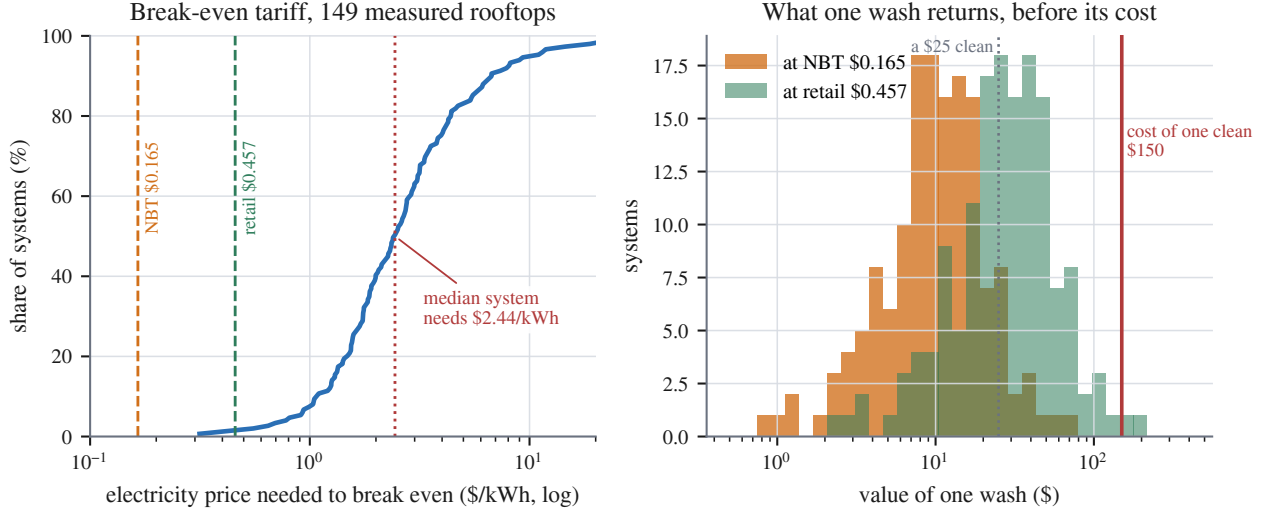


Figure 9: The cleaning-value verdict on 149 measured rooftops, with recovery measured rather than modelled. *Left:* the electricity price each system would need for one professional clean to break even; the median needs \$2.44/kWh against the \$0.165 net-billing blended rate. *Right:* what one wash returns per system at both the blended rate and the full retail offset, against the cost of the service. Both sides of that comparison are per-wash, and the left-hand quantity carries no cleaning assumption. What a wash returns is smaller than any plausible price under either tariff.

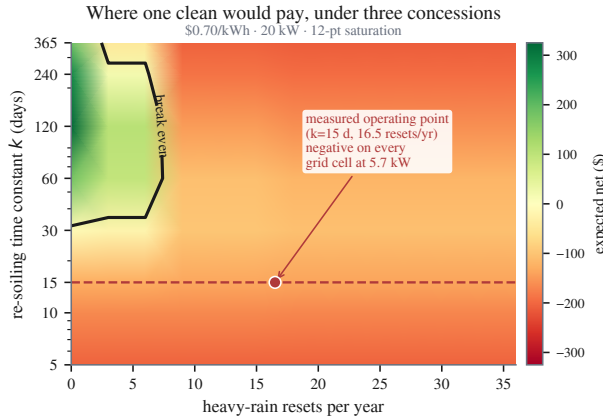


Figure 10: Where one clean would pay, under three concessions stacked in the product’s favour: \$0.70/kWh, a 20 kW roof, and a 12-point saturated soiling level. The measured operating point remains deep in the losing region, and at the measured median system size no grid cell is positive.

unit, and imagery and detector changed together.

5 Discussion

5.1 What the pipeline establishes

Every stage runs on public data, and the answer the last stage returns is *no*. That is the result, not a defect. We reach Mejía and Kleissl [16]’s conclusion for the average California site independently and by a different route, but

our claim is stronger and narrower. On recoverable soiling, at sourced tariffs and a directly measured value per clean, cleaning never pays—not on the 1,865 detected sites in the pilot, and not on the 149 metered rooftops where the array is instrumented and nothing about the loss has to be inferred from imagery. The second population matters more than the first, because it removes detection, area estimation and the risk model from the chain entirely and still returns the same answer.

5.2 The open channel we cannot see

Rain removes dust. Rain does not remove moss, lichen, algae, leaf litter, sap, or built-up droppings, and shade plus moisture encourage them. That channel sits outside the IWSR labels by construction, so nothing in §4.4 speaks to it. We tested the *uniform* persistent-layer hypothesis on a revenue-grade PVDAQ site with 8.08 years of data beside a class-A pyranometer: degradation $-0.076\%/yr$ (95% CI $[-0.408, +0.358]$), standing layer 0.0 points, and the annual maximum soiling ratio pinned at exactly 1.0000 in eight of eight years. That kills the uniform hypothesis. It does not constrain the localised coastal case: that site is a ground mount beside farmland with no canopy, and a whole-array ratio averages away a few bad modules.

The localised loss is wildly nonlinear and *saturating*, not compounding: a shaded cell current-limits its series string until a bypass diode cuts the substring out, so a few square centimetres cost far more than their area fraction and then stop. Modelled at cell granularity [1, 10], four modules each losing one substring cost 7.00%, and a continuous edge band across four portrait modules on string wiring costs 21.00%,

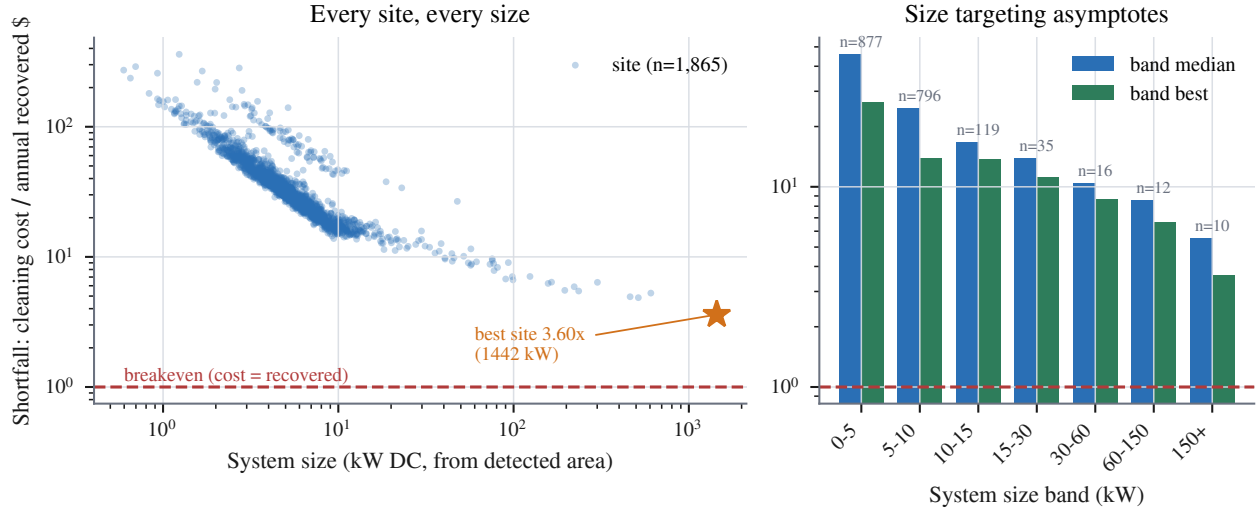


Figure 11: Cleaning cost over annual recovered dollars, all 1,865 sites. No site reaches the breakeven line. Size targeting improves the ratio roughly 8 $\times$ and then asymptotes, because recovered dollars and cleaning cost both scale with panel count.

against breakeven bars of 2.2% to 10.1%. **A wash pays here only if the roof carries continuous edge occlusion across several modules.** That is a specific, falsifiable prediction about what a qualifying roof looks like, and it is exactly what aerial imagery can see and a dust model cannot. Run the moss economics on real per-roof geometry, however, and zero of 2,494 geometry-scored arrays clear a wash, best case losing about \$30, because module-level electronics have been effectively mandated since 2019 and defuse the substring nonlinearity. What remains open is therefore the *measurement*, not the product: nobody has measured biological soiling’s power cost in a Mediterranean climate. Porcar et al. [21] confirmed biofilm in Berkeley but measured no power loss, and our AOI’s growth-weighted time of wetness is 2,444 h/yr against Berkeley’s 2,468, a ratio of 0.99.

5.3 A power result that reorders the obvious experiment

A conventional before/after wash test on a single array saturates at roughly 5% of output minimum detectable effect and *does not improve with more days*, because weather arrives in multi-day blocks and autocorrelation rises at almost exactly the rate at which extra samples would have helped. That is a floor measured on a site with an on-site pyranometer and 893 kW of spatial averaging; a residential roof normalised against satellite irradiance is 1.5 to 2 $\times$ worse. The obvious study, wash 20 random homes and measure the average, is therefore both underpowered and aimed at the wrong estimand. Selecting on visible soiling is not a shortcut; it is what makes the study possible. A split-array wash on one instrumented volunteer roof reaches roughly 1.5% minimum detectable effect in two weeks for about \$150, because weather is common-mode and cancels between halves.

5.4 Limitations

(a) **Label transfer.** We train on instrumented ground stations and deploy on residential rooftops. This is unvalidated and central. (b) **Ranking.** §4.3: valid for level, not for ranking homes. (c) **Level.** The head reads high against every nearby measurement, and the scalar that corrects it treats the symptom. (d) **Recall.** 73.6% on the imaged era and 55.3% across all permit years; the permit denominator is not a census. (e) **Imagery staleness** is distinct from resolution. Any array installed after the flight is invisible regardless of GSD. Our permit audit script still hard-codes a 2022 vintage constant against 2025 imagery, making the imaged-era split *conservative*; this is a known defect, flagged, not yet fixed. (f) **Unsourced constants**, listed in §5.5. (g) **Inter-annotator agreement** was never measured, so the mask ceiling is unknown. (h) **Geography.** Every metered rooftop in the economics is Californian, and the panel labels behind the risk model are 91 % western at station level (Figure 7). The break-even boundary we map— $k \geq 30$ d, at most 15 resets a year, above \$0.20/kWh—is therefore a *prediction* about drier and more expensive markets, not a measurement of them. It is the limitation we would close first. (i) **Label instability.** Per-system soiling labels are not yet well posed: three standard cleaning assumptions place the median annual loss at 2.72, 9.80 and 20.93 % and rank the same systems at ρ between 0.241 and 0.894, with the only plausibly-levelled variant in the minority on ordering. This does not touch the dollar result, which the cancellation of §4.4 makes assumption-free, but it does touch every ranking claim. Any programme that scales per-system label extraction should measure this before trusting a ranking built on it. (j) **Scale.** The free weather archive is the binding constraint on extending beyond one region. In this work it cost roughly four hours of stalled fetching before we noticed that most of what we were

Table 2: Every headline claim, the number behind it, and the strongest objection we know of. Limits are stated here rather than distributed through the text so that a reader can audit the paper’s confidence in one place.

Claim	Number	Strongest limit we know of
Detection is gated and permissive	F_1 0.826 [0.798, 0.853]	One county, one imagery vintage. Permit recall of 73.6% is era-restricted; across all years it is 55.3%.
Our published temporal gate leaked	88.7% of held-out rows kept their own site in training	Measured, not inferred. This one has no serious objection; it is a property of the fold definition.
Station labels cannot rank roofs	0.622 [0.571, 0.670] jointly out-of-site and out-of-year, against 0.710 published	Robust to seed (sd 0.005, 7 seeds), to estimator (logistic regression gives 0.571), and to summary-row weighting (0.594–0.622). The panel label set is 91 % western at station level; the east is censored rows only.
The drop is leakage, not lost data	83% of the 0.0922 fall, 95% CI [52, 99]%	Controlled directly: a size-matched training set that keeps the leak scores 0.6986 against 0.7144 full and 0.6222 clean. The joint fold does cost a median 28.8% of the training rows, and we report that.
Coordinates match the full feature set	0.644 against 0.622	Not a significant win for coordinates either. The claim is the absence of evidence that 38 further features contribute, not evidence of their absence.
Published physics models do no better	Kimber 0.610 [0.566, 0.652], unfitted, against our 0.622	Kimber’s proxy is missing for 50 of 891 rows. The comparison is on ranking only; neither model was built to classify at our threshold.
Does not travel to an unseen region	pooled 0.655, largest region 0.527	Stable across fold geometry (0.638–0.664 for $k = 5 \dots 12$). Regions are k-means cells, not a designed geographic sample.
Per-system labels are necessary, not yet sufficient	levels 2.72 / 9.80 / 20.93 pts across three cleaning assumptions; rankings ρ 0.241, 0.318, 0.894	Our own 0.573 uses <code>perfect_clean</code> , which is the <i>minority</i> ranker, and the only one whose level is plausible. The plausibility check is partly circular: NREL’s IWSR is itself a perfect-clean construction.
The dollar verdict is label-invariant	median \$28.10 per wash under each assumption, all 149 systems (max deviation $\$4 \times 10^{-14}$)	Algebraic, not empirical: annual loss cancels between the dollar figure and the recovery fraction’s denominator. Holds only because we measure the step directly rather than inferring it from a modelled annual loss.
One clean recovers \$28.10 of electricity	median over 505 events on 149 systems; \$150 service	Dry-window events are <i>candidate</i> washes; an inverter reset or a shading change looks the same. Assumes cleaned and un-cleaned trajectories stay parallel. The <i>fraction</i> of annual loss this represents is 0.031–0.217 depending on the cleaning assumption, which is why we report dollars.
Cleaning does not clear	0 of 149 at the \$0.165 blended rate, 2 of 149 at retail; median needs \$2.44/kWh; one wash returns \$28.10 at retail, \$10.14 at the blended rate	Recoverable soiling only; a permanent wash-only layer is excluded by construction. All 149 systems are Californian. Survives the \$150 service floor because the prize is smaller than any plausible price.

requesting was already cached under a different span.

5.5 Unsourced constants, disclosed

The minimum professional service charge (\$150) and the per-panel schedule are our price points, not market quotes, and that one constant decides most residential cases. Published California pricing runs \$5–12 per panel, so our schedule sits

at or below the bottom of the range and currently *flatters* the product. The production anchor of 1,500 kWh/kWp/yr is PVWatts-typical and unmeasured for this AOI. The packing factor (0.90) and midday self-consumption share ($\sigma = 0.30$) are unsourced. The avoided-cost export table is another utility’s standing in for the local one, same methodology, level unverified. The SOMOSclean saturation parameter was fitted against the wrong side of the trajectory: the objective

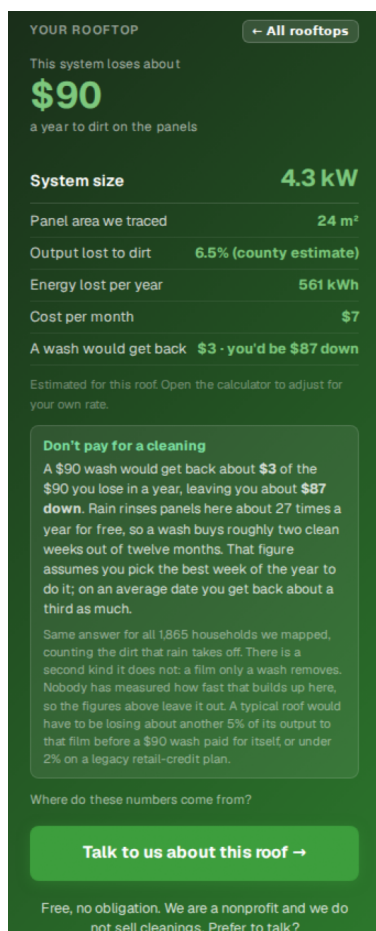


Figure 12: The negative result as a homeowner receives it. The panel reports the loss in dollars, states that a wash leaves this roof about \$87 down, and says so first. It also discloses the channel we cannot measure—“a film only a wash removes... nobody has measured how fast that builds up here”—and that we do not sell cleanings. **This capture predates the level calibration of §4.4:** the 6.5 % shown is above anything the deployed data now contains, whose site-level losses have median 2.77 % and maximum 3.35 %. The overstatement runs in the product’s favour and the panel still says do not pay, which is the property we are evidencing; the arithmetic on a current capture is smaller. The map is cropped out so no individual residence is identifiable.

scored 31 December, mid wet season, against an annual-mean measured IWSR. That does not affect the annual-mean validation, but a refit would not be expected to return 0.08. The net-metering cliff dates are encoded from memory and flagged in code as requiring verification against the regulatory decision.

5.6 Ethics and data provenance

All inputs are public records used within their terms. Addresses are joined from county parcel data to arrays detected in public imagery for the purpose of contacting the property owner; we retain no personally identifying information be-

yond what the mailing requires, and the public dashboard renders array geometry and modelled estimates only. We state on the dashboard that these are modelled estimates from public data, not a professional energy audit and not a guarantee. The deployed panel (Figure 12) does report a percentage and a dollar figure for an individual roof, which is a stronger claim than we would make in text, and we mark the percentage as a county-level estimate rather than a measurement of that roof precisely because we have no production data for it. We regard that labelling as the minimum, not as sufficient: a per-address number carries an authority the underlying model does not earn, and §4.3 is the reason why.

6 Conclusion

A soiling-targeting system can be built end to end on public data, and most of it works. Arrays detect at F_1 0.826 with 73.6% independently measured recall, and per-array roof geometry recovers from public lidar, agreeing with installer filings on the median tilt to 0.03° while disagreeing about the distribution for a reason we can name.

What it cannot do is rank individual roofs, and the validation that suggested otherwise was leaking: folds that hold out a site or a year but never both flatter the model by about nine points. Corrected, the model scores 0.622, two coordinates score 0.644, and an unfitted empirical formula from 2006 scores 0.610. Those three numbers say the same thing from three directions. The cause is structural: public soiling labels are measured at stations, so every roof in a catchment inherits one label and no per-roof feature has anything to fit.

We tested the obvious repair rather than leaving it as an open question, and the answer is partial. Per-system labels derived from public production telemetry *do* carry per-roof signal—0.573 within-cell concordance against a structural ceiling of 0.5—but three standard ways of computing such a label disagree by a factor of eight on the level and by as much as $\rho = 0.241$ on the ordering, and the one whose level is plausible is the one the other two disagree with. The prerequisite for per-roof soiling targeting is therefore a label estimator that is both unbiased for recoverable loss and tight enough to order two roofs in one neighbourhood. That is a measurement problem, and it is the one worth working on.

Whether any of it would matter is settled separately, and more cheaply. On 149 metered rooftops one wash recovers \$28.10 of electricity on the median system at the retail offset, \$10.14 at the net-billing blended rate, against a \$150 service; the median roof needs electricity at \$2.44/kWh to break even. A wash must cost under \$10 before half of these roofs clear at the blended rate. No ranking model, however good, creates value that is not there—which is why we would advise anyone entering this problem to compute the arithmetic before building the pipeline, as we did not.

Two further channels remain genuinely open. The wash-only biological layer our labels exclude by construction may carry measurable power loss in a Mediterranean climate;

nobody has measured it, and an afternoon with one instrumented roof would begin to settle it. And every economic number here comes from Californian systems, so the boundary we map— $k \geq 30$ d, at most 15 rain resets a year, above \$0.20/kWh—is a prediction about drier and more expensive markets rather than a measurement of them.

Reproducibility

Every number in this paper traces to a committed artifact, and the claim is checkable rather than asserted: `paper/verify_numbers.py` reads each headline value out of the JSON that produced it and fails if the rounded form is absent from the manuscript. It is wired into `make paper`, so the PDF cannot be built from stale claims. At submission it passes 30 of 30, and it also fails the build if a *retired* number reappears.

The corrected validation results are produced by a standalone audit that imports nothing from the pipeline it audits and reads the training matrix read-only, so a reader can reproduce the negative results without trusting our code:

Result	Script
Fold taxonomy, leak measurement, 0.622	<code>audit_soiling_validation.py</code>
Reporting-basis reconciliation	<code>audit_reconcile_basis.py</code>
Out-of-region, year axis restored	<code>audit_regional_holdout.py</code>
Seed, estimator, weighting, geometry	<code>audit_robustness.py</code>
Kimber and SOMOSclean baselines	<code>audit_physics_baselines.py</code>
Cleaning events from telemetry	<code>audit_cleaning_intervals.py</code>
Rain reset against manual wash	<code>audit_rain_vs_wash.py</code>
Measured recovery fraction	<code>audit_recovery_empirical.py</code>
Within-cell ranking	<code>audit_per_system_ranking.py</code>
Economics on metered rooftops	<code>audit_real_systems_econ.py</code>
Value per clean, label invariance	<code>audit_value_per_clean.py</code>
Three-assumption label comparison	<code>audit_label_variants.py</code>
Training-size control	<code>audit_trainsize_control.py</code>
Break-even surface	<code>breakeven_surface.py</code>
Detection gate	<code>detect/eval_tile_f1.py</code>
Recall probe	<code>permit_recall_audit.py</code>

All of the above run from cached inputs and need no network. Every figure is emitted by a script under `paper/figures/` that reads those artifacts directly; no figure contains a hand-entered number.

What we got wrong, and how it was caught.

Three results in this paper changed after they had been written down as findings, each because one setting was varied: the recovery fraction turned out to have no single value at all—0.031, 0.063 or 0.217 according to which cleaning

assumption supplies its denominator, which is why this paper reports dollars per wash instead—the within-cell ranking fell from 0.573 to 0.512 on that same choice, and the out-of-region figure moved once the production hyperparameters were actually loaded. None was found by review. We record this because it is the strongest argument we can make for the robustness table in §4.2 and for `audit_robustness.py` existing at all.

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References

- [1] J. W. Bishop. Computer simulation of the effects of electrical mismatches in photovoltaic cell interconnection circuits. *Solar Cells*, 25(1):73–89, 1988. doi: 10.1016/0379-6787(88)90059-2.
- [2] Kyle Bradbury, Raghav Saboo, Timothy L. Johnson, et al. Distributed solar photovoltaic array location and extent dataset for remote sensing object identification. In *Scientific Data*, volume 3, page 160106, 2016. doi: 10.1038/sdata.2016.106.
- [3] California Distributed Generation Statistics. Interconnected project sites, Rule 21, 2026. <https://www.californiadgstats.ca.gov/>.
- [4] California Public Utilities Commission. Net billing tariff, decision 22-12-056, 2022.
- [5] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, pages 213–229, 2020. doi: 10.1007/978-3-030-58452-8_13.
- [6] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- [7] Michael G. Deceglie, Leonardo Micheli, and Matthew Muller. Quantifying soiling loss directly from PV yield. *IEEE Journal of Photovoltaics*, 8(2):547–551, 2018. doi: 10.1109/JPHOTOV.2017.2784682.
- [8] Michael G. Deceglie, Dirk Jordan, Kevin Anderson, et al. RdTools: An open source python library for PV degradation analysis, 2023. <https://github.com/NREL/rdtools>.

- [9] Cameron Gordon. Permissively licensed rooftop array detection at 21 cm, with an executable gate, and per-array roof geometry from public lidar, 2026. Technical report; detection and geometry stages of the pipeline evaluated here. Code and artifacts: <https://github.com/CameronGordon/solar-soiling-detection>.
- [10] William F. Holmgren, Clifford W. Hansen, and Mark A. Mikofski. pvlib python: a python package for modeling solar energy systems. *Journal of Open Source Software*, 3(29):884, 2018. doi: 10.21105/joss.00884.
- [11] International Energy Agency. Electricity 2026: Analysis and forecast to 2028, 2026.
- [12] International Renewable Energy Agency. Renewable capacity statistics 2026, 2026.
- [13] Adrienne Kimber, Lawrence Mitchell, Steve Nogradi, and Howard Wenger. The effect of soiling on large grid-connected photovoltaic systems in California and the Southwest region of the United States. In *IEEE 4th World Conference on Photovoltaic Energy Conference*, pages 2391–2395, 2006. doi: 10.1109/WCPEC.2006.279690.
- [14] Alexander Kirillov, Eric Mintun, Nikhila Ravi, et al. Segment anything. *arXiv preprint arXiv:2304.02643*, 2023.
- [15] Kevin Mayer, Benjamin Rausch, Marie-Louise Arlt, Gunther Gust, Zhecheng Wang, Dirk Neumann, and Ram Rajagopal. 3D-PV-Locator: Large-scale detection of rooftop-mounted photovoltaic systems in 3D. *Applied Energy*, 310:118469, 2022. doi: 10.1016/j.apenergy.2021.118469.
- [16] Felipe A. Mejía and Jan Kleissl. Soiling losses for solar photovoltaic systems in California. *Solar Energy*, 95: 357–363, 2013. doi: 10.1016/j.solener.2013.06.028.
- [17] Leonardo Micheli, Michael G. Deceglie, and Matthew Muller. Predicting photovoltaic soiling losses using environmental parameters: An update. *Progress in Photovoltaics: Research and Applications*, 29(2):230–240, 2021. doi: 10.1002/pip.3352.
- [18] Leonardo Micheli, Michael G. Deceglie, and Matthew Muller. PV soiling map of the United States. Technical report, National Renewable Energy Laboratory, 2023. <https://www.nrel.gov/pv/soiling.html>.
- [19] National Renewable Energy Laboratory. PVDAQ: Photovoltaic data acquisition, OEDI data lake, 2024. <https://openei.org/wiki/PVDAQ>.
- [20] Open-Meteo. Historical weather and air quality API, 2024. <https://open-meteo.com/>.
- [21] Manuel Porcar, Katherine B. Louie, Suzanne M. Kosina, et al. Microbial ecology on solar panels in Berkeley, California, United States. *Frontiers in Microbiology*, 9:3043, 2018. doi: 10.3389/fmicb.2018.03043.
- [22] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, et al. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- [23] Roboflow. RF-DETR: A real-time transformer-based object detector, 2025. Apache-2.0. <https://github.com/roboflow/rf-detr>.
- [24] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32*, 2019.
- [25] Solar Energy Industries Association. Solar market insight report, 2026.
- [26] U.S. Energy Information Administration. Short-term energy outlook: Small-scale solar generation, 2023.
- [27] USDA Farm Production and Conservation. National agriculture imagery program (NAIP), 2023. <https://naip-usdaonline.hub.arcgis.com/>.
- [28] Jiafan Yu, Zhecheng Wang, Arun Majumdar, and Ram Rajagopal. DeepSolar: A machine learning framework to efficiently construct a solar deployment database in the United States. *Joule*, 2(12):2605–2617, 2018. doi: 10.1016/j.joule.2018.11.021.
- [29] Daniele Zanaga, Ruben Van De Kerchove, Dirk Daems, et al. ESA WorldCover 10 m 2021 v200. 2022. doi: 10.5281/zenodo.7254221.